

Adapting the MEE Pipeline to PUNCH Coronagraph Data

Centroid detection, Gaia validation, and what the noise floor taught us

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Executive Summary

Over the past few weeks I adapted our MEE2024 Tab 1 centroid pipeline to PUNCH Level-3 white-light coronagraph images. PUNCH turned out to be different enough from eclipse data that I went through three full pipeline iterations before getting a clean result — the corona itself generates thousands of false detections, so I ultimately masked the inner field and validated everything that remained against the Gaia star catalog using PUNCH's mission pointing solution.

The pipeline works: of 4,343 detected centroids in the final run, 86% correspond to real catalog stars. But the same analysis quantified a hard limitation — at PUNCH's plate scale ($\sim 81''/\text{pixel}$), the gravitational deflection signal for even our closest detectable star is roughly 1,000x below our measured position noise. Moving closer to the Sun doesn't fix this: at PUNCH's resolution, no observable radius puts the signal above the noise.

The value of this exercise is that the machinery — centroid finding, catalog matching, and the deflection sensitivity check — is now built and validated, and the numbers tell us exactly what kind of instrument it needs next: a finer plate scale that reaches closer to the Sun. For future runs I also plan a visual-first step (confirming stars by eye before trusting automated matches), a check I skipped on this first pass.

1. Data & Setup

- Instrument:** PUNCH (Polarimeter to Unify the Corona and Heliosphere), Level-3 white light.
- Frame:** 2026-02-19 00:16 UTC, 4096 x 4096 pixels, Sun centered at pixel (2048, 2048).
- Plate scale:** ~ 81 arcsec/pixel — about 44x coarser than our eclipse data ($\sim 1.85''/\text{px}$); $1 R_{\odot} \approx 12$ px.
- The challenge:** the corona fills the inner $\sim 20\%$ of the frame with bright, lumpy structure that the centroid finder happily mistakes for stars.

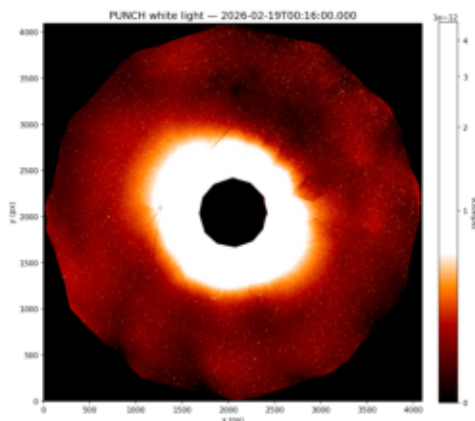


Figure 1. The 2026-02-19 PUNCH white-light frame. The Sun (black disk) is surrounded by the bright corona; real stars are faint points in the outer field. Everything inside the corona is a minefield for automated star detection.

2. What I Did — Three Iterations to a Clean Detection

Run	Threshold	Corona handling	Centroids	Outcome
01	$\sigma = 6$	none (MEE's blob mask never fires on PUNCH)	9,255	mostly corona junk
02	$\sigma = 10$	none	4,824	still corona-dominated
03	$\sigma = 10$	circle mask, $r < 1000$ px	4,343	clean — used for analysis

MEE's built-in saturation mask assumes an over-exposed Sun; PUNCH's Sun is a dark hole, so that mask never activated. For run 03 I wrote my own pre-processing step: every pixel within 1,000 px of Sun-center is replaced with a single flat sky value (the median of a thin ring just outside the mask), blanking 18.7% of the image. This removes the corona's false bumps before the centroid finder ever sees them.

One important consequence: at 12 px per solar radius, a 1,000 px mask means the nearest star we can possibly detect sits at ~ 83 solar radii. That inner limit in everything below is self-imposed by the mask — not a physical property of the data.

2.1 Validating against Gaia

A detected centroid counts as a match if it lands within 2 pixels of where the Gaia catalog says a star should be (projected through PUNCH's pointing solution). Two numbers summarize the result:

- Purity = 86.3% — of the 4,343 things I flagged, 3,749 are real stars. The detector rarely cries wolf.
- Completeness = 44.8% — of the 8,560 catalog stars bright enough to be in the usable field, I recovered about 45%. The fainter half stay below PUNCH's noise.

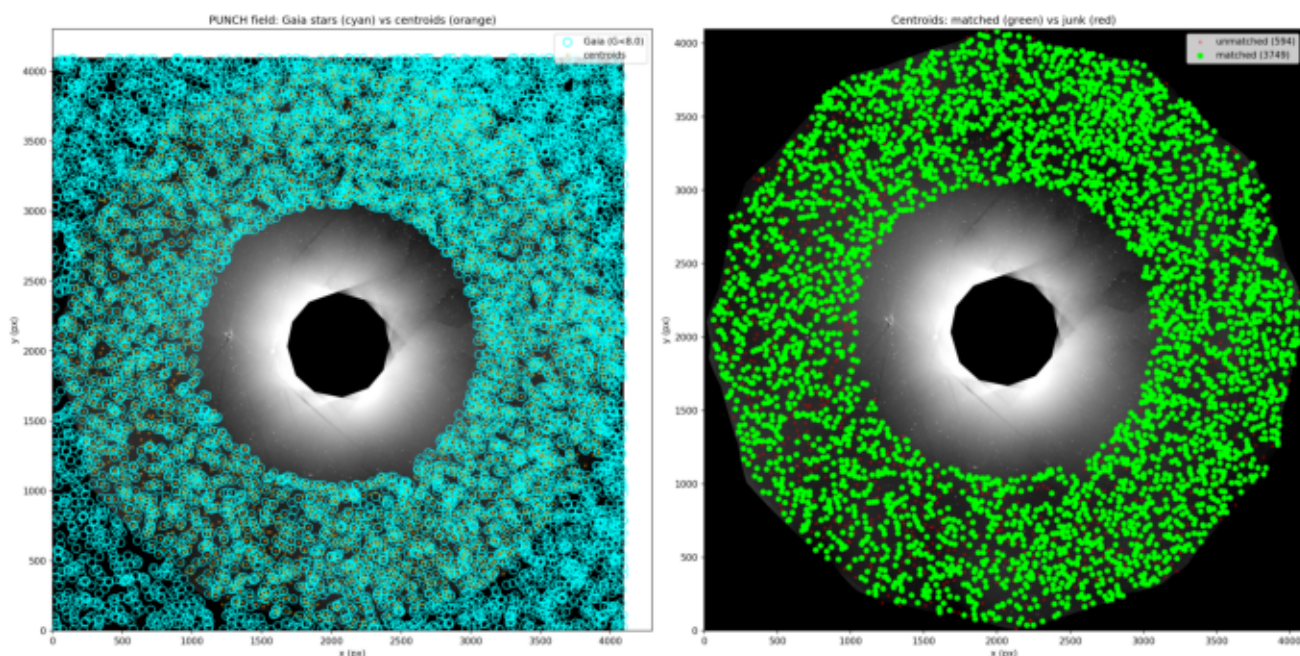


Figure 2. Run 03 validation. Left: detected centroids (cyan) on the masked PUNCH frame. Right: the Gaia truth set — green points are catalog stars recovered by the pipeline. The empty central region is the $r < 1000$ px corona mask.

3. The Limiting Result — Deflection Signal vs. Noise Floor

Closest detectable star:	83.7 solar radii from Sun center (set by the corona mask)
Predicted Einstein deflection there:	0.021 arcsec $\approx$ 0.0003 pixels
Measured position noise:	22.6 arcsec $\approx$ 0.28 pixels (median centroid-to-Gaia residual)
The gap:	signal is $\sim 1,000\times$ below the noise

The natural question is whether detecting stars closer to the Sun would close the gap. It would not. Deflection grows as $1.75''/r$, so to lift the signal above even a tenth of a pixel at PUNCH's scale, a star would need to sit at $r < 0.2$ solar radii — inside the Sun. The limitation is not where our stars are; it is that 81" pixels are too coarse for this measurement at any radius.

That is a useful thing to know precisely, and it points directly at what would work: the same pipeline pointed at a coronagraph with a finer plate scale, which can both resolve smaller shifts and see stars much nearer the limb.

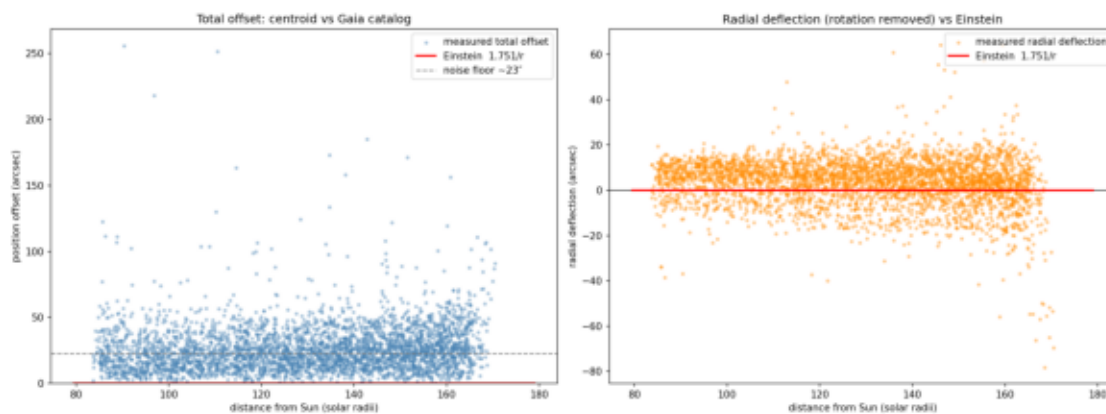


Figure 3. Measured star offsets vs. distance from the Sun (left: total offset, right: radial component). The red curve is the predicted Einstein deflection ($1.75''/r$) — three orders of magnitude below the measurement noise at every distance PUNCH can see. The dashed gray line is the $\sim 23''$ noise floor.

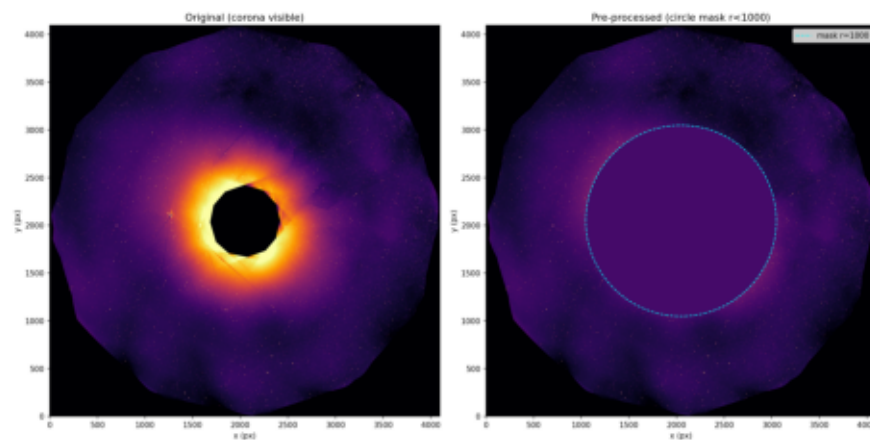


Figure 4. The corona mask that made run 03 work. Left: original frame. Right: the pre-processed input actually fed to the centroid finder — everything inside $r < 1000$ px (dashed circle) is flattened to a single sky value.

4. Refined Approach & Next Steps

4.1 A visual-first check

One lesson from the first pass: I trusted the automated brightest-detections list without first confirming by eye which of them were actually stars. A conversation with Henry Throop (NASA) reinforced a better habit — inspect the frame visually and confirm star-like sources before matching. (For this I use DS9, a standard free astronomy viewer for FITS images that lets you zoom and adjust the brightness stretch.)

As a first application, I built a "top-20" comparison: the 20 brightest detected centroids plotted against the 20 brightest predicted Gaia stars, with no automatic matching — pure visual inspection. It was immediately informative: several of the brightest detections near the Sun have no catalog counterpart at all (they are corona structure, not stars), while the nearest visually confirmed real star sits at ~ 98 solar radii, with other confirmed matches out to ~ 160 .

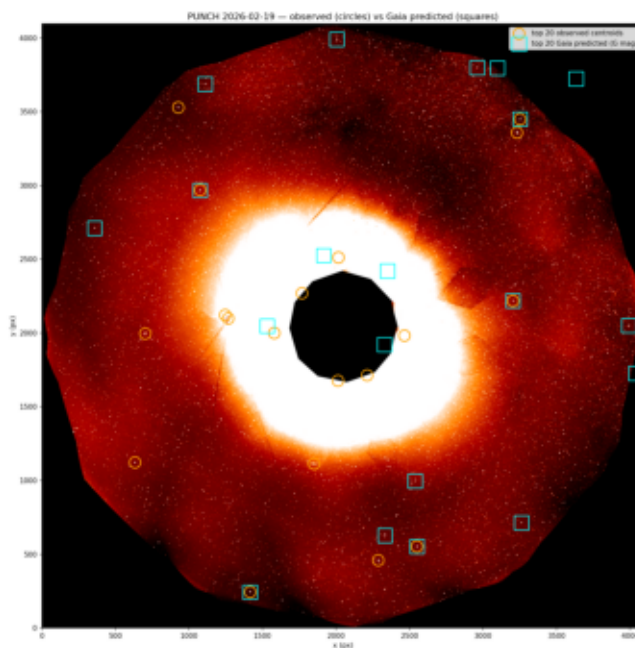


Figure 5. Top-20 brightest detections (orange circles) vs. top-20 predicted Gaia stars (cyan squares). Where a circle has no square, the "star" is corona. The closest genuine match, at roughly the 3 o'clock position, lies at ~ 98 solar radii.

4.2 A working principle I've adopted

For anything astrometric, I now treat the native mission FITS as the only source of truth. Display products (JP2s, PNGs) are stretched, 8-bit, and lossy-compressed — I verified that a JP2 re-wrapped as FITS carries only 256 distinct pixel values versus thousands in the real data — so they cannot be used for sub-pixel position work, no matter the file extension.

Next steps

- Apply the visual-first workflow to establish a trusted star list per frame.
- Repeat the brightest-N analysis across multiple PUNCH dates: same stars, many epochs — this validates the method and pins down our real error budget (even though PUNCH itself cannot yield L).
- In parallel, I'm looking for a suitable dataset from a finer-scale coronagraph (the STEREO instruments, at $\sim 15''/\text{px}$ and reaching within a few solar radii of the limb, are the leading candidates). If I can source a good science-grade frame, that's where this pipeline gets pointed next.